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		ACCEPTANCE TEST PROCEDURE		DOC CODE ATP	PART NUMBER 4556 109 201 15		
SWITCH FILTER ASSEMBLY (RFU MK-1B)		02	5253601	ISSUE	SH No.	PREPD.	DATE
		01	5185822	02	2	ANIL KUMAR	22.11.13
		00	ORIGINAL	HELD IN	No. OF SH	APPRD.	DATE
		ISS No.	CN No.	CD-1	37	DAMODARAN V	22.11.13
7.1 PHYSICAL INSPECTION REMARKS 22 7.2 FUNCTIONAL TEST REPORT 23 ANNEXURE- A: PRODUCTION QUALITY TESTING 33 ANNEXURE- B : APPROVED QUALITY PLAN 37							

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		ISS No.	CN No.	CD-1	37	DAMODARAN V	22.11.13
LIST OF ABBREVATIONS: ATR ATTEN BPF COC CW DLVA EQUI FOD MDI PP PSD SOLT SW VSWR Acceptance Test Report Attenuation Band Pass Filter Certificate Of Conformance Continuous Wave Detector Logarithmic Video Amplifier Equaliser Foreign Object Debris Master Drawing Index Purchase Part Power Spectral Density Short Open Load Through SWitch Voltage Standing Wave Ratio							

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		ISS No.	CN No.	CD-1	37	DAMODARAN V	22.11.13
1.0 RELATED DOCUMENTS: PP Drawing number: 4556 109 201 15 A3.							
2.0 FUNCTIONAL DESCRIPTION: SFA operates in the frequency range of 2 - 18 GHz. SFA provides an accurate amplitude information of the received RF pulses. The unit has got two input ports for RF IN and BIT. The unit has got an RF output port. The unit houses in a 25 PIN D-type connector thru which power supply and control signals are fed to the unit and Log VID OUT is taken out. The functional block diagram is shown in Fig.1 Front end RF preamplification is provided to achieve a high sensitivity of -65dBm for amplitude measurement. The objective of SFA is to provide an accurate amplitude information on the received RF pulses. In addition, a RF output is available with a nominal RF gain 39±3.25dB for further processing.							
2.1 BRIEF DESCRIPTION OF SFA: The Switch Filter Assembly, AKON Model A20-ML146, is a miniature, lightweight and rugged front-end RF unit. It covers the frequency range 2.0 to 18.0 GHz and uses MIC and MMIC modules. The assembly comprises of four blocks: • Input Module with amplifier and SP4T switch • Band Pass Filters • Attenuator – Equalizer • Detector Log Video Amplifier (DLVA) The front-end is designed for a minimum of –71 dBm Tangential Signal Sensitivity and the log range extends from –65.0 to 0.0 dBm. The frequency coverage of 2.0 to 18.0 GHz is in four bands, 2.0 to 6.0 GHz, 6.0 to 10.0 GHz, 10.0 to 14.0 GHz, and 14.0 to 18.0 GHz. The band selection is done through four differential band select signals. The assembly provides a video output with a rise time of less than 40 nS. The DLVA used is CW immune. CW signal Detection is possible by chopping the signal at a predetermined rate by toggling the enable port of the SP2T switch in the Input Module. An auxiliary output port, before detection, with a minimum gain of 36 dB, has been provided for Instantaneous Frequency Measurement.							

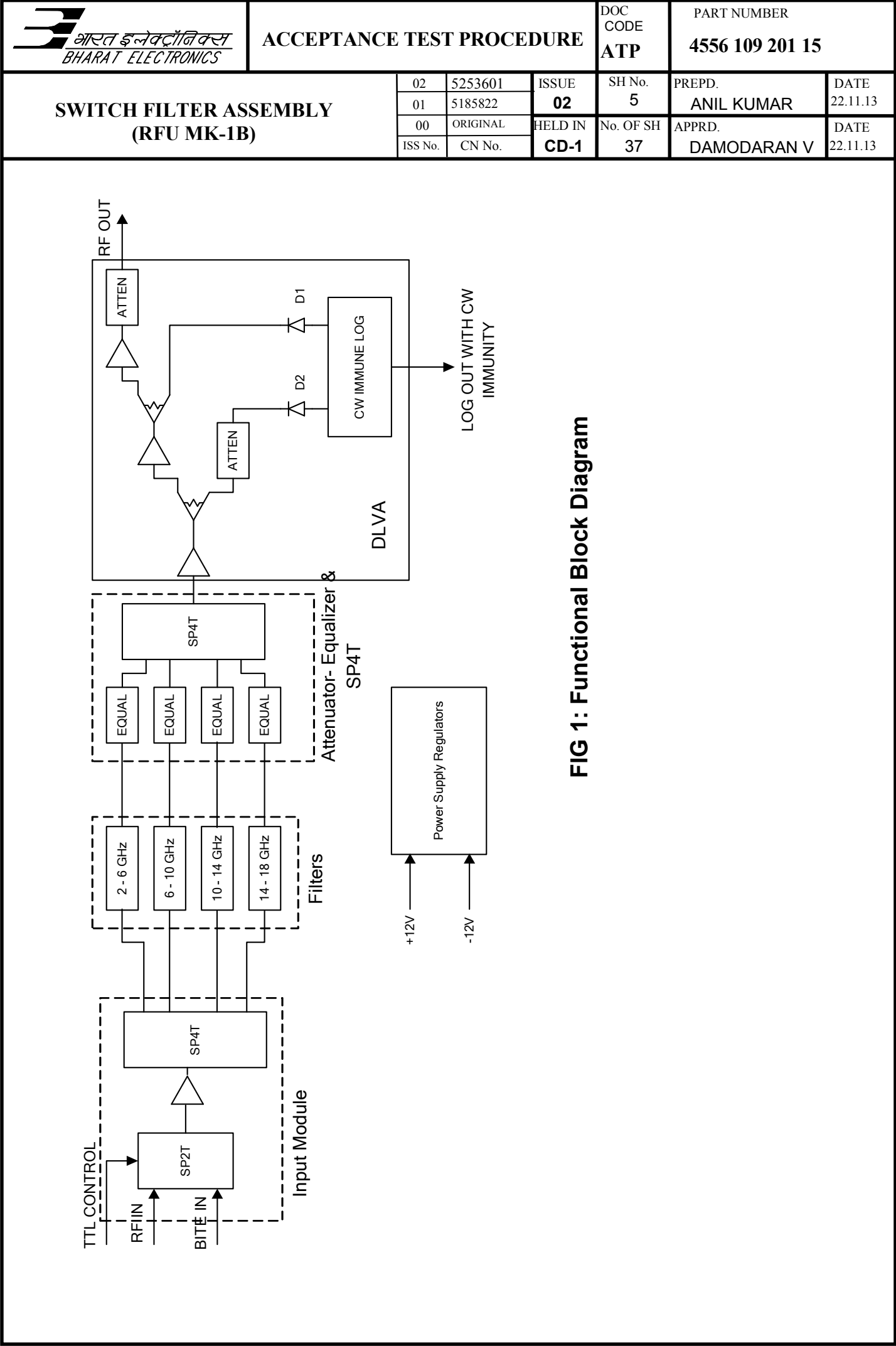


FIG 1: Functional Block Diagram

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3.0 SPECIFICATION:							
3.1 ELECTRICAL SPECIFICATION:							
SL NO	PARAMETER			SPECIFICATIONS			
1	Frequency			2 To 18 GHz covered in 4 Bands; 2-6 GHz, 6-10 GHz, 10-14 GHz, 14-18 GHz			
2	Pass Band Sensitivity(At Video Port)						
2.1	At 15 dB Output SNR			Better Than -65 dBm			
2.2	T.S.S			Better Than -71 dBm			
3	Linear Dynamic Log Range			65 dB (-65 dBm To 0 dBm)			
4	Absolute accuracy of power measurement in the centre of the Sub-Bands at 25°c ambient at the mounting surface pass band around a straight line with co-ordinates 500mv at -65 dBm and 4400 mV at 0 dBm with a slope of 60 mV/dB.			± 2.0 dB			
5	Absolute accuracy of power over frequency (Frequency Flatness) and temperature around a straight line with co-ordinates 500 mV at -65 dBm and 4400 mV at 0 dBm with a slope of 60 mV/dB			± 3.0 dB			
6	Channel to Channel isolation for any frequency in the Pass Band, for an input such that the RF Output is not compressed			65 dB Min			
7	Pass Band/ Stop Band Details						
	Band	Pass band	Stop Band				
	Band I:	2-6 GHz	DC- 1.7 GHz 6.9-18 GHz				
	Band II:	6-10 GHz	DC -5.22 GHz 11.5-18GHz				
	Band III:	10-14 GHz	DC- 8.69 GHz 16.1-18.0 GHz				
	Band IV:	14-18 GHz	DC- 12.1 GHz				
8	Antenna Port To Bit Port Isolation			Better than 65dB			
9	Absolute Max. Input Level						
	For RF Input			+30 dBm			
	For BIT Input			+20 dBm			
10	Input VSWR			3.0: 1			
11	Output RF Port						
11.1	Small Signal Gain Within Pass band			39dB ±3 dB			
11.2	Maximum Saturated Output Power At RF O/P Port			+13 dBm			
11.3	Output Broad-Band Noise Power in each Band			-30 dBm Nom. For All Bands -29 dBm Max			
11.4	Output SNR At RF O/P Port At -65 dBm			2dB (Min)			

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SL NO	PARAMETER	SPECIFICATIONS					
	Input						
11.5	Harmonics @ 0 dBm RF I/P	-8 dBc Minimum					
12	RF Input Pulse Width	100 ns To 14 μ s					
13	Video Port	(Measured With 12 Mtrs, 95 Ohms Raychem Cable Part No: 9532A1114-9 Or Equivalent)					
13.1	Output Baseline Noise Measured using a true RMS Volt meter equivalent To HP 3400 A With 10 MHz Video Bandwidth	50 mV RMS Max. 30 mV RMS Nominal.					
13.2	Output D.C Offset with RF I/P terminated To 50 Ω	0 mV $\pm$ 50 mV					
	C.W. Immunity	C.W signal less than -53dBm shall produce less than 200mV peak offset plus noise relative to ground					
13.3	C.W Chopping:	When the unit is used in the chopping mode with 1 μ s wide pulse at 10 μ s PRI, the maximum offset plus noise shall be less than 200mV peak up to 0dBm C.W level					
14	Video Output (Peak Relative To Ground)						
14.1	-65 dBm Input At Antenna Port	500 mV $\pm$ 180 mV					
14.2	0 dBm Input At Antenna Port	4400 mV $\pm$ 180 mV					
14.3	Log Slope For Accuracy Window	60 mV / dB Fixed					
14.4	Log Linearity Over The Full Frequency And Temperature Range At 60 mV/dB Slope	$\pm$ 1.5 dB					
15	Pulse Characteristics						
15.1	Rise Time (10% To 90%)	40 ns Maximum					
15.2	Setting Time (10% To 0.5 dB of Final Value)	60 ns Maximum					
15.3	Recovery Time (1 dB Below Peak To A Level Produced By -65 dBm Signal)	350 ns Maximum					
15.4	Falling Edge Re – Entrance	The Positive Over Shoot, Excursion and Ringing at the end of the falling end shall remain under threshold. i.e., 200 mV for pulse width variation of 100 ns To 14 μ s					
16	Video Output	Short Circuit Protected					
17	Switching Spike Leakage At Video Output Without RF Input	Should Be Less Than 200mV					
18	Delay Between 50% TTL Enable Signal to 3 dB below (180 mV) peak of video shall be	210 ns $\pm$ 20 ns					
19	Maximum Delay Variation in a given band between any two units for 50% RF to 3 dB below (180 mV) peak of video pulse for up to 10 dB differential input power level shall be	20 ns Maximum					

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<table><tr><th>SL NO</th><th>PARAMETER</th><th>SPECIFICATIONS</th></tr><tr><td>20</td><td>Duty cycle variation for maintaining the peak of pulsed signal relative to ground with in the accuracy specification of Para 1.1.15</td><td></td></tr><tr><td>20.1</td><td>Pulse Width of 14 μs</td><td>10%</td></tr><tr><td>20.2</td><td>For Pulse Width Between 14 μs To 100 ns</td><td>40% if the min. PRI Between Two Pulses is greater than 2.5 μs 4% at 2.5 μs PRI</td></tr><tr><td>21</td><td>A. Band Selection Control (A0,A1,A2)</td><td>Through Differential Signal Lines As Per RS 422 Standard, To Be Terminated With 470Ω</td></tr><tr><td></td><td>B. Enable Signal (E)</td><td>Through Differential Signal Lines As Per RS 422 Standard, To Be Terminated With 120Ω</td></tr><tr><td>22</td><td>DC Power (Maximum)</td><td>+12V@ 1050 mA -12 V @ 250 mA</td></tr><tr><td>23</td><td>The Ripple and Noise on the input supply shall be less than 100 mV p-p in DC to 10 MHz range. The input voltage tolerance is ± 2%</td><td></td></tr><tr><td>24</td><td>Connectors</td><td></td></tr><tr><td>24.1</td><td>RF In Port (J4)</td><td>SMA Female (M39012/60-3001)</td></tr><tr><td>24.2</td><td>BIT Port (J3)</td><td>SMA Female (M39012/60-3001)</td></tr><tr><td>24.3</td><td>RF Out Port (J5)</td><td>SMA Female (M39012/60-3001)</td></tr><tr><td>24.4</td><td>DC Power & Signals Video Port (J2)</td><td>25 Pin D Type Sub Miniature P/N: M24308/3-3</td></tr></table>										SL NO	PARAMETER	SPECIFICATIONS	20	Duty cycle variation for maintaining the peak of pulsed signal relative to ground with in the accuracy specification of Para 1.1.15		20.1	Pulse Width of 14 μs	10%	20.2	For Pulse Width Between 14 μs To 100 ns	40% if the min. PRI Between Two Pulses is greater than 2.5 μs 4% at 2.5 μs PRI	21	A. Band Selection Control (A0,A1,A2)	Through Differential Signal Lines As Per RS 422 Standard, To Be Terminated With 470Ω		B. Enable Signal (E)	Through Differential Signal Lines As Per RS 422 Standard, To Be Terminated With 120Ω	22	DC Power (Maximum)	+12V@ 1050 mA -12 V @ 250 mA	23	The Ripple and Noise on the input supply shall be less than 100 mV p-p in DC to 10 MHz range. The input voltage tolerance is ± 2%		24	Connectors		24.1	RF In Port (J4)	SMA Female (M39012/60-3001)	24.2	BIT Port (J3)	SMA Female (M39012/60-3001)	24.3	RF Out Port (J5)	SMA Female (M39012/60-3001)	24.4	DC Power & Signals Video Port (J2)	25 Pin D Type Sub Miniature P/N: M24308/3-3																									
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5	/A1	12	-12V RTN	19	NC																																																																				
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<table><tr><th colspan="4">INPUTS</th><th colspan="3">SFA OPERATING MODE</th></tr><tr><th>E</th><th>A2</th><th>A1</th><th>A0</th><th>BAND</th><th>MODE</th><th>PORT</th></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>I</td><td rowspan="4">CW</td><td rowspan="7">ANTENNA*</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>II</td></tr><tr><td>0</td><td>0</td><td>1</td><td>0</td><td>III</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>IV</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>I</td><td rowspan="3">PULSE*</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>II*</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>III</td></tr></table>										INPUTS				SFA OPERATING MODE			E	A2	A1	A0	BAND	MODE	PORT	0	0	0	1	I	CW	ANTENNA*	0	0	0	0	II	0	0	1	0	III	0	0	1	1	IV	1	0	0	1	I	PULSE*	1	0	0	0	II*	1	0	1	0	III												
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SL NO	PARAMETER					SPECIFICATIONS	
1	0	1	1	IV	CW	BITPORT	
0	1	0	1	I			
0	1	0	0	II			
0	1	1	0	III			
0	1	1	1	IV			
1	1	0	1	I	PULSE		
1	1	0	0	II			
1	1	1	0	III			
1	1	1	1	IV			
*This is the default operating mode of the SFA							
26	Temperature range						
	Operating					-40 ⁰ c to +71 ⁰ c, ambient	
	Storage					-54 ⁰ c to +85 ⁰ c, ambient	
27	Reliability MTBF					7000hrs.	

3.2 MECHANICAL SPECIFICATION

SI No	Parameter	Specification
1	Dimensions	191.01mm X 83.21mm X 31.2mm
2	Weight	1.4 kg Max

4.0 TEST PROCEDURE

4.1 PHYSICAL INSPECTION

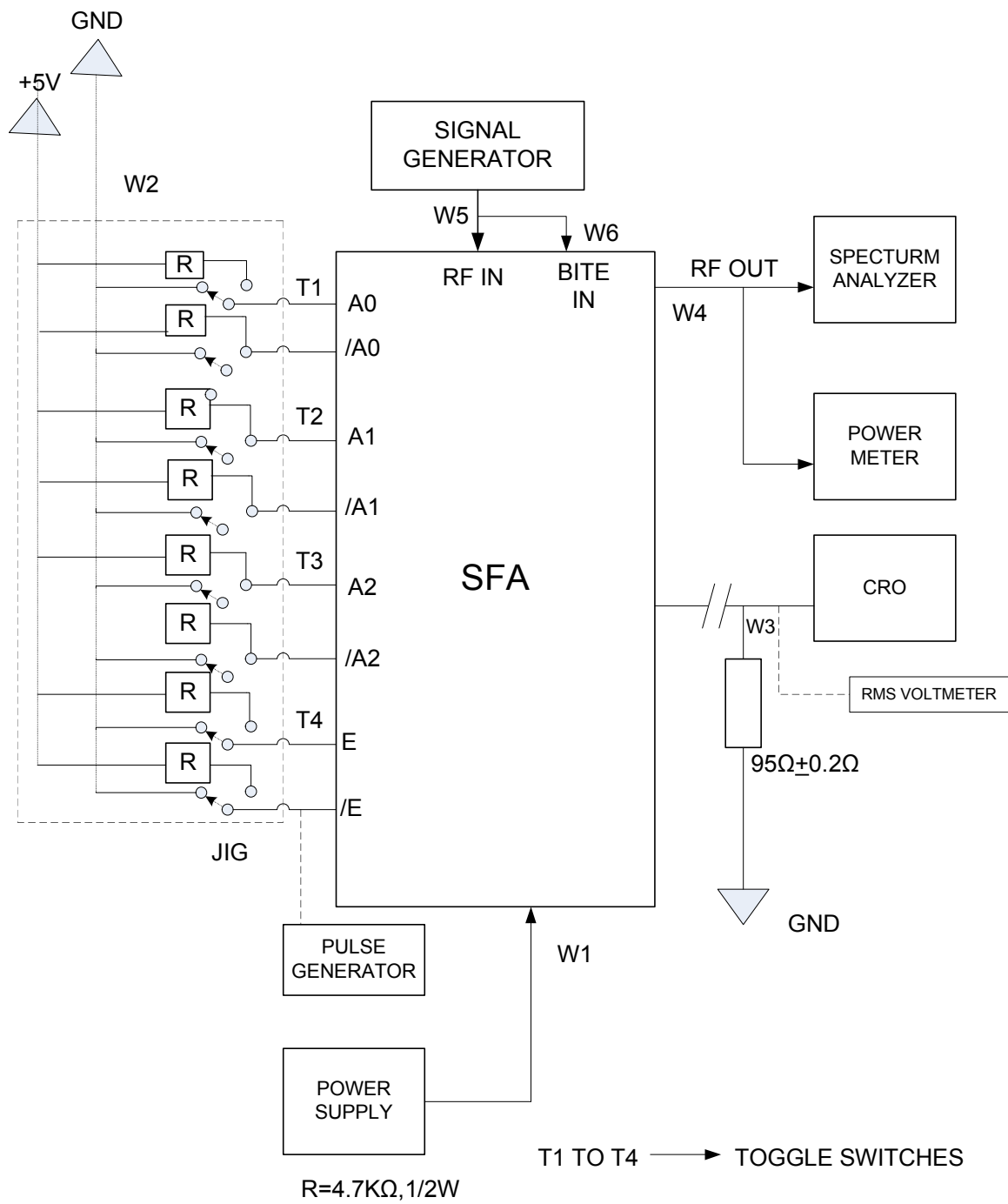
Verify requirements as per the below list and enter the results in Test report 7.1 on the offered unit. Physical inspection of the unit to be done as per BEL drawing no: 4556 109 201 15 PP as per latest configuration control document when required.

- Details of the Unit viz. SI No, Year code and BEL P/N.
- QM/EW&A accepted report of the Unit.
- OEM ATR for the Unit.
- Verify for FOD – solder splash, Paint or lose items etc.
- Verify overall dimensions of the Unit as per PP drawing.
- Verify dimensions and centre distance(CD) of the mounting holes of the Unit with Fixture no: 8137 164 500 54
- Verify weight of the Unit.
- Verify COC of the OEM.

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	ISS No.	CN No.	CD-1	37	DAMODARAN V	22.11.13				
4.2 TEST INSTRUMENTS AND ACCESSORIES:										
1. Multi output Power supply -1 no.										
<table><tr><td>+12 V :</td><td>2 A</td></tr><tr><td>-12 V :</td><td>0.5 A</td></tr></table>							+12 V :	2 A	-12 V :	0.5 A
+12 V :	2 A									
-12 V :	0.5 A									
2. Oscilloscope 500 MHz, 1 GS/s, Storage type, Tektronics TDS 724C or equivalent -1 no.										
3. Signal generator 0.1GHz to 20 GHz, HP 83731B or equivalent -1 no.										
4. Universal power meter, Gigatronics 8652A or equivalent -1 no.										
5. Spectrum Analyzer, HP E4407B or equivalent -1 no.										
6. Low Loss RF cable, S130BFSS10032, SEMFLEX or equivalent -1 no.										
7. Low Loss RF Cable, S130BSS10230, SEMFLEX or equivalent -1 no.										
8. Vector Network Analyzer, Agilent E8362B or Equivalent -1 no.										
9. 50 Ω Terminator, OR equivalent - no.										
10. Cable, Video (4879 100 601 40 A3) -12 Mts.										
11. Pulse Generator HP8110 or equivalent -1no.										
12. Test Jig No: T-RF/002/003/004 -1no.										
13. RMS volt meter Model: HP 3400B or equivalent -1no.										

 भारत इलेक्ट्रॉनिक्स BHARAT ELECTRONICS		ACCEPTANCE TEST PROCEDURE			DOC CODE ATP	PART NUMBER 4556 109 201 15		
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4.3 TEST SETUP



*12 METER VIDEO CABLE

FIG-2 : SFA TEST SET UP

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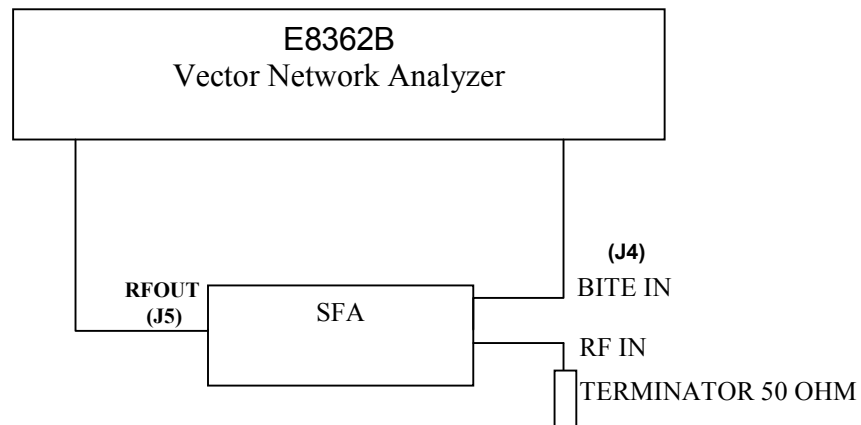


FIG 3: Test set-up for Gain Measurement

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		01	5185822	02	13	ANIL KUMAR	22.11.13
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4.3.1 CONNECTION DETAILS:

LOOM	SIGNAL NAME POWER SUPPLY	SFA CONNECTOR – J2 PIN NO.	Type of Wire
W1	+12	1	
	-12	2	
	GND	11,12	

LOOM	CONTROL SIGNALS	SFA CONNECTOR – J2 PIN NO.	Type of Wire
W2	Ao	15	7/0.2 WIRE
	A1	14	
	A2	16	
	E	25	
	/A0	4	
	/A1	5	
	/A2	6	
	/E	7	

LOOM	Signal Name	SFA CONNECTOR J2 PIN NO	Video Cable P/N
W3	VID OUT	23	4871 100 601 40 A3
	VID RTN	24	4871 100 601 40 A3

LOOM	Signal Name	SFA CONNECTOR	Type of Wire
W4	RF OUT	J5	S130BFSS10230
W5	RF IN	J4	S130BFSS10032
W6	BITE IN	J3	S130BFSS10032

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				ISS No.	CN No.				
4.3.2 BAND SELECTION CONTROL SIGNALS:									
INPUTS				SFA Operating Mode					
E	A2	A1	A0	BAND	MODE	PORT			
0	0	0	1	I	CW	ANTENNA			
0	0	0	0	II					
0	0	1	0	III					
0	0	1	1	IV					
1	0	0	1	I	PULSE				
1	0	0	0	II					
1	0	1	0	III					
1	0	1	1	IV					
0	1	0	1	I	CW	BIT PORT			
0	1	0	0	II					
0	1	1	0	III					
0	1	1	1	IV					
1	1	0	1	I	PULSE				
1	1	0	0	II					
1	1	1	0	III					
1	1	1	1	IV					

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4.4 FUNCTIONAL TEST PROCEDURE:

Note: terminate all unused ports with 50 Ohms terminator.

4.4.1 POWER CONSUMPTION MEASUREMENT:

SUPPLY VOLTAGE	MAXIMUM CURRENT LIMIT
+ 12 V	1050 mA
- 12 V	250 mA

1. Make the inter connections as per the set-up shown in Fig : 2.
2. Switch on the Power supply.
3. Check all the Voltages.
4. Measure the current in the +12 V, -12 V points at quiescent conditions of SFA. & record the results in the test report SI.No.: 1
5. The measured values should be as per the specification below.

4.4.2 RF MEASUREMENTS:

4.4.2.1 GAIN MEASUREMENT:

➤ PASS BAND FREQUENCIES

Spec: RF Gain: 35.75 dB min, 42.25 dB max.

1. Make the interconnections as per the set-up shown in the Fig.3.
2. Calibrate the Network Analyser System for Transmission measurement using “thru” calibration.
3. Attenuators may be used to get proper RF I/P to the SFA. Appropriate correction factors to be applied for the same.
4. Refer Table-1 for selection of PORT's and Bands.
5. Choosing E & A2 provides MODE & PORT selection.
6. Choosing A0 & A1 provides BAND selection.
7. Feed RF I/P of -50dBm to RF I/P port of SFA in CW mode.
8. Make the Sweep measurements from 2GHz to 18GHz by selecting different bands

Band 1 : 2.0 to 6 GHz

Band 2 : 6 to 10 GHz

Band 3 : 10 to 14 GHz

Band 4 : 14 to 18 GHz.

9. When the swept measurements are completed plot/print the band response.
10. The measured values should be within the specification.

1. Terminate RF I/P and BITE ports with 50Ω.
2. Follow Table-1 for band selection and select Band-1.
3. Measure the O/P power level using Power Meter.
4. Record the results in Test Report.
5. Repeat above for Band-2, Band-3, Band-4 record the measured values.
6. The measured values should be within the specifications shown above.

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4.4.2.4 SNR MEASUREMENT

Spec:

Output SNR at Band-1:	2dB Min.
Output SNR at Band-2:	2dB Min.
Output SNR at Band-3:	2dB Min.
Output SNR at Band-4:	2dB Min.

1. Set the power level of input signal to -65 dBm & frequency to 4GHz and give it to RF IN Port of RFU.
2. Refer Table-1 for band selection and select the antenna port.
3. Measure the power level of output signal (S) in dBm.
4. Switch off the input signal and measure the output power level. This gives the Noise level (N) in dBm.
5. The difference between S & N (S-N) gives SNR.
6. Repeat the same for 8GHz, 12GHz, 16GHz and record the measured values as per Test report.
7. The measured values should be within the specifications shown above.

4.4.2.5 HARMONICS MEASUREMENT

Spec: Harmonics : Better than 8dBc.

1. With the same set-up as in fig 2, replace the Power Meter with Spectrum Analyser.
2. Refer Table-1 for band selection and select the antenna port.
3. Set the frequency of input signal to 2 GHz and Power level to 0 dBm.
4. Record the harmonics level in Test Report.
5. Repeat the above for other frequencies of 4GHz, 6GHz, 8GHz & record the measured values.
6. The measured values should be within the specifications shown above.

4.4.2.6 CHANNEL TO CHANNEL ISOLATION MEASUREMENT:

Spec: For any frequency in the pass bands the output measured in the stop band shall be > 63 dB.

1. With the same set-up as in fig 2, replace the Power Meter with Spectrum Analyser.
2. Refer Table-1 for band selection and select the antenna port .
3. Set the input signal power level to -35dBm and frequency to 5.22 GHz.
4. Set the centre frequency of Spectrum Analyser to 5.22 GHz and measure the amplitude and make it as reference (0dBm) using marker.
5. Switch to other Bands individually by referring to Table-1 and record the measured values in the Test report.
6. The measured values should be within the specifications.
7. Repeat the above for other frequencies 6.9GHz, 8.69GHz, 11.5GHz, 11.8GHz, 16.1GHz, and 18GHz.

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4.4.2.7 ANTENNA TO BITE PORT ISOLATION MEASUREMENT:

Spec: Isolation > 65dB.

1. Make the interconnections as per test set-up shown in Fig: 2.
2. Terminate the RF OUT Port with 50Ω and connect BITE port to Spectrum Analyser.
3. Follow Table-1 and select antenna port.
4. Feed the signal of 4GHz and 0dBm at RF input port of SFA.
5. Measure the output at BITE port.
6. Record the measure values as per Test Report. Measured value should be within the specification.

4.4.3 VIDEO PORT MEASUREMENTS:

4.4.3.1 TRANSFER RESPONSE MEASUREMENT:

Spec:

At band centers: (Frequencies@4GHz, 8GHz, 12GHz, 16GHz)	At band edges: (Frequencies@2GHz, 6GHz, 10GHz, 14GHz, 18 GHz)
V _{out} @-65 dBm: 500 mV ± 135mV V _{out} @ 0 dBm: 4400 mV ± 135mV. Accuracy: 4.5dB P-P Max	V _{out} @ -65 dBm: 500 mV ± 195mV V _{out} @ 0 dBm: 4400 mV ± 195mV. Accuracy: 6.5dB P-P Max

4.4.3.1.1 C.W. MODE:

➤ **PASS BAND FREQUENCIES**

1. Make the interconnections as per set up shown in Fig: 2 & Select CW Mode.
2. Feed a TTL pulse from pulse generator of PW = 1μs, PRI = 10 μs to Enable Pin (J2-25).
3. Set the frequency of input RF signal using Signal Generator to 2GHz frequency.
4. Vary the input power from -65 dBm to 0dBm as per test report and measure the output using CRO.
5. Record the measured values as per the Test Report.
6. Calculate the accuracy and record the result in Test Report.

$$A = 500 \text{ mV} + \text{INT} [(B+65) * 60], B = \text{I/P Power in dBm.}$$
$$\text{Accuracy} = [V_{\text{out}}-A] / 60 \text{ dB, Where } V_{\text{out}} \text{ is in mV}$$

7. The measured values should be within the specifications.
8. Repeat the above for all pass band frequencies of 4GHz, 8GHz, 12GHz, 16GHz 18GHz.
9. The measured P-P accuracy should be within the specifications when the I/P power varies from –65 dBm to 0 dBm

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4.4.3.1.2 PULSE MODE:

1. Feed the pulse from signal Generator with following pulse parameters
Delay = 0 ns, PW = 1 μ s, PRI = 10 μ s.
2. Repeat the procedure as in CW Mode all frequencies.
3. Record the measured values in the test report.
4. The measured values should be within the specifications.

4.4.3.2 SENSITIVITY MEASUREMENT:

Spec: SNR ≥ 15 dB

1. Make the interconnections as per set up shown in Fig: 2.
2. Feed the pulse from signal Generator with following pulse parameters
Delay = 0 ns, PW = 1 μ s, PRI = 10 μ s.
3. Feed a 2GHz signal @ -65dBm to Antenna Port.
4. Follow Table-1 for band selection and select the antenna port.
5. Measure the VID OUT signal (S) from CRO.
6. Switch off the input signal and measure the Video output (N) from RMS Volt meter.
7. Calculate the sensitivity as $20 * \text{Log} (S/N)$. Record the measured values as per the Test Report.
8. Repeat the above for other frequencies of 4GHz, 8GHz, 12GHz, 16GHz, 18GHz & record the measured values.
9. Repeat the above measurements for BITE port. Record the values in Test Report.
10. The measured values should be within the specifications.

4.4.3.3 DUTY CYCLE AND RISE TIME MEASUREMENT

Spec: At band center: 4GHz, 8GHz, 12GHz, and 16GHz

$$V_{out} @ -65 \text{ dBm}: 500 \text{ mV} \pm 135 \text{ V}$$

$V_{out}@ 0 \text{ dBm}: 4400 \text{ mV} \pm 135\text{V}$

Rise time ≤ 40 ns

1. Make the interconnections as per test set up shown in Fig: 2.
2. Feed the pulse from signal Generator with following pulse parameters
Duty cycle = 10%, PW = 14 μ s, PRI = 140 μ s.
3. Set input RF signal power level to 0dBm & Frequency to 4GHz and select Band-1by following the Table-1.
4. Measure the VID OUT using Oscilloscope.
5. Measure the VID OUT for the input power of -65 dBm also.
6. Record these results in test report
7. Repeat the above measurements for PW=14 μ s, PRI = 35 μ s Duty Cycle=40%,and record the values as per Test Report.
8. The measured values should be within the specifications.
9. Make similar settings as above and feed the input power level of -45dBm to measure the rise time of output video pulse. Record the rise time for both the settings of duty cycle.
10. Repeat the above measurements and record the results for input frequency of 8GHz, 12GHz and 16GHz.

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4.4.3.4VIDEO NOISE MEASUREMENT : Spec: ≤ 50mV. 1. Make the interconnections as per test set up shown in Fig: 2. 2. Switch off the Signal Generator. 3. Refer table-1 and select Band-1. 4. Measure the VID OUT signal. This gives the Video Noise. 5. Repeat the same for Band-2, Band-3, and Band-4 and record the measured values as per Test Result. 6. The measured values should be within the specifications.						

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5.0 ESS TESTS: ESS test report supplied by the vendor will be verified by TS/EW.																																																	
SL NO	PARAMETER	SPECIFICATION																																															
1	Stabilization Bake	Without applying any DC power the unit would be at +125°C for 96 Hours																																															
2	Thermal shock	As per MIL-STD-202, Method 107 Temperatures: -54°C and +125 °C Dwell time at each Temperature: 30 minutes Transfer time from one temperature to the other temperature: 15 seconds Max. The above constitutes 1-cycle. Conduct the Performance check at the end of 10 th and 20 th cycles, total 20 cycles.																																															
3	Random vibration(operating test)	As per MIL-STD-883, Method2026, Test condition E, 16.4 grms from 50Hz to 2000Hz. Test for 10 minutes in the Z-axis(perpendicular to the mounting axis) and 5 minutes in the other 2 Axes																																															
4	Mechanical shock	As per MIL-STD-202F, Method 213B 100G, 6 milliseconds, ½ sine 5-Shock cycles Y-1 Axis(Perpendicular to Mounting axis of LRU)																																															
5	Temperature cycling	The unit shall be subjected to 24 temperature cycles. Each temperature cycle is described below and takes 3.5Hrs Cycle description 1st and 24th cycles: <table><tr><th>Temp</th><th>Time</th><th>DC Power</th></tr><tr><td>-40°C Cold soak</td><td>30min</td><td>OFF</td></tr><tr><td>-40°C wake up & warm up</td><td>5min</td><td>ON</td></tr><tr><td>-40°C operational</td><td>25min</td><td>ON</td></tr><tr><td>Transition to +71°C</td><td>15min</td><td>ON</td></tr><tr><td>Soak at +71°C</td><td>60mins</td><td>ON</td></tr><tr><td>+71°C operational</td><td>25min</td><td>ON</td></tr><tr><td>Total dwell time at +71°C</td><td>2.5Hrs</td><td>ON</td></tr><tr><td>Transition to -40°C</td><td>15min Max</td><td>OFF</td></tr></table> Cycle description 2nd through 23rd cycles: <table><tr><th>Temp</th><th>Time</th><th>DC power</th></tr><tr><td>-40°C cold soak</td><td>30min</td><td>OFF</td></tr><tr><td>Transition to +85°C</td><td>15min</td><td>ON</td></tr><tr><td>dwell time at +85°C</td><td>2.5Hrs</td><td>ON</td></tr><tr><td>Transition to -40°C</td><td>15min Max</td><td>OFF</td></tr></table>						Temp	Time	DC Power	-40°C Cold soak	30min	OFF	-40°C wake up & warm up	5min	ON	-40°C operational	25min	ON	Transition to +71°C	15min	ON	Soak at +71°C	60mins	ON	+71°C operational	25min	ON	Total dwell time at +71°C	2.5Hrs	ON	Transition to -40°C	15min Max	OFF	Temp	Time	DC power	-40°C cold soak	30min	OFF	Transition to +85°C	15min	ON	dwell time at +85°C	2.5Hrs	ON	Transition to -40°C	15min Max	OFF
Temp	Time	DC Power																																															
-40°C Cold soak	30min	OFF																																															
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dwell time at +85°C	2.5Hrs	ON																																															
Transition to -40°C	15min Max	OFF																																															
6	RF Burn-In	With DC Power ON, at 0dBm input, At +85°C for 48Hours minimum																																															

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6.0 COMPATIBILITY CHECK:

RFU shall be checked in TARANG MK1B system integration stage by giving frequencies 5GHz, 8GHz, 12GHz & 15GHz at power level of -30 dBm. If there is valid symbol as per PFM, the RFU is considered as compatible with TARANG MK1B system.

7.0 TEST REPORT:

Tests Conducted for the Contract No / Sale order (Ref No): _____

Drawing No: 4556 109 201 15 Serial No: Date:

7.1 PHYSICAL INSPCTION REPORT:

Activity			Observation	Acceptance by BEL Rep	Acceptance by ORDAQA Rep
Initial physical inspection					
a) Check for over all dimensions (mm) (Tolerances as per PP drawing)	L	191.01			
	B	88.9			
	H	31.2			
b) Following dimension to be verified with Inspection Jig. Tool No.: 8137 164 500 54					
Reference Zone in drawing: 4556 109 201 15 PP	Dimensions in mm (Tolerance as per drawing 4556 109 201 15 PP)				
B1	86.87				
B2	86.87				
A2	80.14				
c) Following dimension to be verified with drawing 4556 109 201 15 PP					
Reference Zone in PP drawing	Dimensions in mm (Tolerance as per drawing)				
B1	8.64				
A3	22.9				
A3	8.38				
d) Check marking details as per drawing 4556 109 201 15 PP					
e) Visual inspection					
1.SMA female connector:		3 Nos			
2. D-type connector - 25 pin (As per MIL-C-24308):		1No			
3.Screen Print Texts (As per drawing 4556 109 201 15)					

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Activity	Observation	Acceptance by BEL Rep	Acceptance by ORDAQA Rep
f) Following to be verified with respect to CoC.			
1. Paint 26231 texture Grey per MIL-C-85285			
2. Mounting surface: Electro less Nickel Per MIL-C-26074E,Class 3,Grade B			
3.Date code and SI No.			
g) FOD check			
h) Weight (1.4Kg Max)			

7.2 FUNCTIONAL TEST REPORT

Tests Conducted for the Contract No / Sale order (Ref No):

Drawing No.: 4556 109 201 15SI. No:Date:

7.2.1 TEST INSTRUMENTS USED:

Sl. No.	Instrument description	Model No.	Sl. No	Calibration status
1				
2				
3				
4				
5				
6				
7				
8				

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7.2.2 TEST REPORT FOR LAB TEST:							
SI No	TEST	Test Parameter	Specified Value Of Current.	Measured Value (mA)	Test Result (PASS/FAIL)	Remarks	
1	6.1 Power Consumption Measurement	Current for following Voltages					
		+12V	1050mA Max.				
		-12V	250 mA Max.				
SI.No	TEST	Test Parameter	Specified Value (dB)	Test Result (PASS/FAIL)	Remarks		
2	4.4.2.1 RF Gain Measurement (Pass Band Frequencies)	RF Gain @ -50 dBm			Refer gain plot		
		Band-1	35.75dB Min, 42.25dB Max.				
		Band-2					
		Band-3					
		Band-4					

		ACCEPTANCE TEST PROCEDURE			DOC CODE ATP	PART NUMBER 4556 109 201 15																																																																																														
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<table><tr><th>SL No</th><th>TEST</th><th>Test Parameter</th><th>Specified Value (dBm)</th><th>Measure d Value (dBm)</th><th>Test Result (Pass/Fail)</th><th>Remarks</th></tr><tr><td rowspan="8">3</td><td rowspan="8">4.4.2.2 Maximum O/P Power Measurement</td><td>Maximum Power at RF Port.@ 0dBm I/P</td><td></td><td></td><td></td><td></td></tr><tr><td>4GHz</td><td>13dBm Max..</td><td></td><td></td><td></td></tr><tr><td>8GHz</td><td>13dBm Max..</td><td></td><td></td><td></td></tr><tr><td>12GHz</td><td>13dBm Max..</td><td></td><td></td><td></td></tr><tr><td>16GHz</td><td>13dBm Max..</td><td></td><td></td><td></td></tr><tr><td>Maximum Power at BITE Port. @ 0dBm I/P</td><td></td><td></td><td></td><td></td></tr><tr><td>4GHz</td><td>13dBm Max..</td><td></td><td></td><td></td></tr><tr><td>8GHz</td><td>13dBm Max..</td><td></td><td></td><td></td></tr><tr><td>12GHz</td><td>13dBm Max..</td><td></td><td></td><td></td></tr><tr><td>16GHz</td><td>13dBm Max..</td><td></td><td></td><td></td></tr></table> <table><tr><th>SL No</th><th>TEST</th><th>Test Parameter</th><th>Specified Value(dBm)</th><th>Measured Value(dBm)</th><th>Test Result (PASS/FAIL)</th><th>Remarks</th></tr><tr><td rowspan="5">4</td><td rowspan="5">4.4.2.3 Noise Power Measureme nt</td><td>Broad Band Noise</td><td></td><td></td><td></td><td></td></tr><tr><td>Band 1</td><td>-29 dBm Max.</td><td></td><td></td><td></td></tr><tr><td>Band 2</td><td>-29 dBm Max.</td><td></td><td></td><td></td></tr><tr><td>Band 3</td><td>-29 dBm Max.</td><td></td><td></td><td></td></tr><tr><td>Band 4</td><td>-29 dBm Max.</td><td></td><td></td><td></td></tr></table>								SL No	TEST	Test Parameter	Specified Value (dBm)	Measure d Value (dBm)	Test Result (Pass/Fail)	Remarks	3	4.4.2.2 Maximum O/P Power Measurement	Maximum Power at RF Port.@ 0dBm I/P					4GHz	13dBm Max..				8GHz	13dBm Max..				12GHz	13dBm Max..				16GHz	13dBm Max..				Maximum Power at BITE Port. @ 0dBm I/P					4GHz	13dBm Max..				8GHz	13dBm Max..				12GHz	13dBm Max..				16GHz	13dBm Max..				SL No	TEST	Test Parameter	Specified Value(dBm)	Measured Value(dBm)	Test Result (PASS/FAIL)	Remarks	4	4.4.2.3 Noise Power Measureme nt	Broad Band Noise					Band 1	-29 dBm Max.				Band 2	-29 dBm Max.				Band 3	-29 dBm Max.				Band 4	-29 dBm Max.			
SL No	TEST	Test Parameter	Specified Value (dBm)	Measure d Value (dBm)	Test Result (Pass/Fail)	Remarks																																																																																														
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		12GHz	13dBm Max..																																																																																																	
		16GHz	13dBm Max..																																																																																																	
		Maximum Power at BITE Port. @ 0dBm I/P																																																																																																		
		4GHz	13dBm Max..																																																																																																	
		8GHz	13dBm Max..																																																																																																	
12GHz	13dBm Max..																																																																																																			
16GHz	13dBm Max..																																																																																																			
SL No	TEST	Test Parameter	Specified Value(dBm)	Measured Value(dBm)	Test Result (PASS/FAIL)	Remarks																																																																																														
4	4.4.2.3 Noise Power Measureme nt	Broad Band Noise																																																																																																		
		Band 1	-29 dBm Max.																																																																																																	
		Band 2	-29 dBm Max.																																																																																																	
		Band 3	-29 dBm Max.																																																																																																	
		Band 4	-29 dBm Max.																																																																																																	

		ACCEPTANCE TEST PROCEDURE			DOC CODE ATP	PART NUMBER 4556 109 201 15			
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SL No	TEST	Test Parameter	Specified Value(dBm)	Measured Value(dBm)	Test Result (PASS/FAIL)	Remarks
	4.4.2.4 SNR Measurement	SNR @ -65dBm I/P 4GH 8GHz 12GHz 16GHz	≥ 2dB ≥ 2dB ≥ 2dB ≥ 2dB			

SL No	TEST	Test Parameter	Specified Value(dBm)	Measured Value (dBm)	Test Result (PASS/FAIL)	Remarks
5	4.4.2.5 Harmonics Measurement	Harmonics @ 0dBm I/P BAND 1 2GHz 4GHz BAND 2 6GHz 8GHz	Better than 8dBc			

7. CHANNEL TO CHANNEL ISOLATION (Para 4.4.2.6)

Input @ -35dBmSpecified Value: > 63dB

Frequency (GHz)	5.22	6.9	8.69	Test Result Pass/fail	Remarks
Band	Band 1	Band 2	Band 2		
Band 1					
Band 2					
Band 3					
Band 4					

		ACCEPTANCE TEST PROCEDURE			DOC CODE ATP	PART NUMBER 4556 109 201 15																																													
SWITCH FILTER ASSEMBLY (RFU MK-1B)				02	5253601	ISSUE	SH No.	PREPD.	DATE																																										
				01	5185822	02	27	ANIL KUMAR	22.11.13																																										
				00	ORIGINAL	HELD IN	No. OF SH	APPRD.	DATE																																										
				ISS No.	CN No.	CD-1	37	DAMODARAN V	22.11.13																																										
7. CHANNEL TO CHANNEL ISOLATION (Para 4.4.2.6) Input @ -35dBmSpecified Value: > 63dB <table><thead><tr><th>Freque ncy (GHz)</th><th>11.5</th><th>11.8</th><th>16.1</th><th>18</th><th>Test Result Pass/fail</th><th>Remarks</th></tr></thead><tbody><tr><td>Band</td><td>Band 3</td><td>Band 3</td><td>Band 4</td><td>Band 4</td><td></td><td></td></tr><tr><td>Band 1</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Band 2</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Band 3</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>Band 4</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></tbody></table>										Freque ncy (GHz)	11.5	11.8	16.1	18	Test Result Pass/fail	Remarks	Band	Band 3	Band 3	Band 4	Band 4			Band 1							Band 2							Band 3							Band 4						
Freque ncy (GHz)	11.5	11.8	16.1	18	Test Result Pass/fail	Remarks																																													
Band	Band 3	Band 3	Band 4	Band 4																																															
Band 1																																																			
Band 2																																																			
Band 3																																																			
Band 4																																																			
SI No	TEST	Test Parameter	Specified Value(dBm)	Measured Value (dBm)	Test Result (PASS/FAIL)	Remarks																																													
8	4.4.2.7 Antenna to BITE port Isolation Measureme nt	Isolation @ 0dBm	> 65dBm																																																

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9a.TRANSFER RESPONSE (Para 4.4.3.1.1) CW MODE

Spec:

At band centers: (Frequencies@4GHz, 8GHz, 12GHz, 16GHz)	At band edges: (Frequencies@2GHz, 6GHz, 10GHz, 14GHz, 18 GHz)
V _{out} @ -65 dBm: 500 mV ± 135mV V _{out} @ 0 dBm: 4400 mV ± 135mV. Accuracy: 4.5dB P-P Max	V _{out} @ -65 dBm: 500 mV ± 195mV V _{out} @ 0 dBm: 4400 mV ± 195mV. Accuracy: 6.5dB P-P Max

Input Power (dBm)	-20		-10		0		Test Result Pass/Fail	Remarks
	Vout (mV)	Accuracy(dB)	Vout (mV)	Accuracy (dB)	Vout (mV)	Accuracy(dB)		
Freq. (GHz)								
2								
4								
6								
8								
10								
12								
14								
16								
18								

9b. TRANSFER RESPONSE (Para 4.4.3.1.2) PULSE MODE

Spec: At band centers:

(Frequencies@4GHz, 8GHz, 12GHz, 16GHz)

At band edges:

(Frequencies@2GHz, 6GHz, 10GHz, 14GHz, 18 GHz)

V_{out} @ -65 dBm: 500 mV $\pm$ 135mV V_{out} @ 0 dBm: 4400 mV $\pm$ 135mV. Accuracy: 4.5dB P-P Max	V_{out} @ -65 dBm: 500 mV $\pm$ 195mV V_{out} @ 0 dBm: 4400 mV $\pm$ 195mV. Accuracy: 6.5dB P-P Max
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[illegible]

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9(b)TRANSFER RESPONSE (Para 4.4.3.1.2) PULSE MODE

Spec:

At band centers: (Frequencies@4GHz, 8GHz, 12GHz, 16GHz)				At band edges: (Frequencies@2GHz, 6GHz, 10GHz, 14GHz, 18 GHz)			
V _{out} @ -65 dBm: 500 mV ± 135mV V _{out} @ 0 dBm: 4400 mV ± 135mV. Accuracy: 4.5dB P-P Max				V _{out} @ -65 dBm: 500 mV ± 195mV V _{out} @ 0 dBm: 4400 mV ± 195mV. Accuracy: 6.5dB P-P Max			

Input Power (dBm)	-20		-10		0		Test Result Pass/Fail	Remarks
	Vout (mV)	Accura cy(dB)	Vout (mV)	Accura cy(dB)	Vout (mV)	Accurac y(dB)		
Freq. (GHz)								
2								
4								
6								
8								
10								
12								
14								
16								
18								

Sl.No.	TEST	Test Parameter	Specified Value	Measur ed Value	Test Result (PASS/FAIL)	Remarks
10a	4.4.3.2 Sensitivity Measurement At ANTENNA Port.	SNR @- 65dBm I/P 2GHz	≥ 15dB			
		4GHz	≥ 15dB			
		8GHz	≥ 15dB			
		12GHz	≥ 15dB			
		16GHz	≥ 15dB			
		18GHz	≥ 15dB			
10b	4.4.3.2 Sensitivity Measurement At BITE Port.	SNR @- 65dBm I/P 2GHz	≥ 15dB			
		4GHz	≥ 15dB			
		8GHz	≥ 15dB			

		ACCEPTANCE TEST PROCEDURE			DOC CODE ATP	PART NUMBER 4556 109 201 15	
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Sl.No.	TEST	Test Parameter	Specified Value	Measur ed Value	Test Result (PASS/FAIL)	Remarks
		12GHz	≥ 15dB			
		16GHz	≥ 15dB			
		18GHz	≥ 15dB			

11. DUTY CYCLE VARIATIOIN MEASUREMENT (Para 4.4.3.3)

Spec:

At band centers: (Frequencies@4GHz, 8GHz, 12GHz, 16GHz)

V_{out} @ -65 dBm: 500 mV ± 135mV

V_{out}@ 0 dBm: 4400 mV ± 135mV.

Accuracy: 4.5dB P-P Max

	PW= 14μS,PRI=140μS Duty cycle = 10%		PW= 14μS,PRI=35μS Duty cycle = 40%		Test Result Pass/Fail	Remarks
BAND	Video out @0dBm I/P(mV)	Video out @-65dBm I/P(mV)	Video out@0dBm I/P(mV)	Video out @-65dBm I/P(mV)		
BAND1						
BAND2						
BAND 3						
BAND 4						

12. RISE TIME MEASUREMENT (Para 4.4.3.3)

Specified Value: ≤ 40nS Max.

	@ PW=4μS,PRI=140μS Duty cycle = 10% I/P	@ PW= 14μS,PRI=35μS Duty cycle = 40% I/P	Test Result Pass/Fail	Remarks
BAND	RISE TIME (nS) @ -45dBm I/P	RISE TIME (nS) @-45dBm I/P		
BAND1				
BAND 2				
BAND 3				
BAND 4				

Sl.No	TEST	Test Parameter	Specified Value	Measured Value	Test Result (PASS/FAIL)	Remarks
13	4.4.3.4 Video Noise Measureme nt	Video Noise Band-1	$\leq 50\text{mV}$			
		Band-2	$\leq 50\text{mV}$			
		Band-3	$\leq 50\text{mV}$			
		Band-4	$\leq 50\text{mV}$			

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PROFILE 1

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A4.1.1 OPERATIONAL TESTING Transfer Response test as per table below to be carried out by feeding the pulsed RF Signal from signal generator to RF IN Port with following pulse parameters Delay=0 ns, PW=1 µs, PRI=10 µs.													
Frequency (GHz) Input Power(dBm)		Band 1		Band 1		Band 2		Band 3		Band 4		Spec: 2600mV±180mV	
		2		4		8		12		16		Test Result Pass/Fail	
		Vout (mV)	Acc (dB)	Vout (mV)	Acc (dB)	Vout (mV)	Acc (dB)	Vout (mV)	Acc (dB)	Vout (mV)	Acc (dB)		
-30dBm													
A4.2 Low Temperature-Storage and Operation:													
A4.2.1 Storage SFA unit shall be subjected to Low Temperature Storage Test in accordance With the following parameters and Profile. Chamber Temperature : -54°C. Duration : 4 Hrs.													
A4.2.2 Operating SFA shall be subjected to a Low Temperature Operational Test in accordance with the following parameters. Chamber Temperature : -40 °C Duration : 4 Hours Continuously ON													
A4.2.3 OPERATIONAL TESTING Transfer Response test as per Para A4.1.1 to be carried out.													
A4.3 High Temperature - Storage & Operation													
A3.3.1 Storage SFA unit shall be subjected to High Temperature Storage Test in accordance With the following parameters. Chamber Temperature : +105°C. Duration : 12 Hrs.													

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A3.3.2 Operation

SFA shall be subjected to a High Temperature Operational Test in accordance With the following parameters.

Chamber Temperature : +71 °C
Duration : 4 Hours Continuously ON

Note: The amount of time required to ramp the temperature is dependent on the test facility change rate. The volume of the chamber shall be at least 10 times the DUT.A ground has to be extended from DUT to test setup.

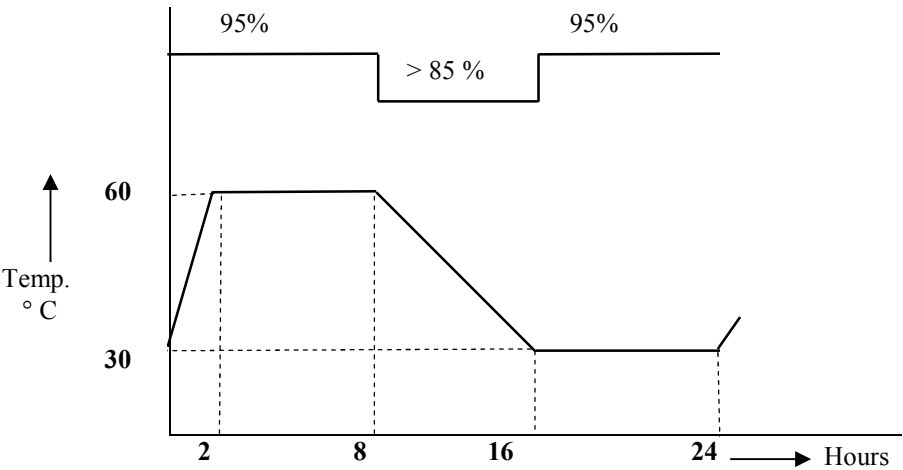
A4.3.3 OPERATIONAL TESTING

Transfer Response test as per Para A4.1.1 to be carried out.

A4.4 Humidity (Non Operational)

SFA shall be subjected to a Damp Heat (Humidity) Test consisting of 5 Cycles in accordance with the following parameters and Profile

Humidity : 85 to 95% R.H
Temperature : +30 to +60 ° C
Cycle Period : 24 Hrs.
No.of Cycles : 5



DURATION: 5 Cycles

Note:

1. The amount of time required to ramp the temperature is dependent on the test facility change rate.
2. At the end unit shall undergo Physical Inspection.

		ACCEPTANCE TEST PROCEDURE			DOC CODE ATP	PART NUMBER 4556 109 201 15																																																	
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		01	5185822	02	36	ANIL KUMAR	22.11.13																																																
		00	ORIGINAL	HELD IN	No. OF SH	APPRD.	DATE																																																
		ISS No.	CN No.	CD-1	37	DAMODARAN V	22.11.13																																																
A4.5 Altitude (Operational) SFA shall be subjected to altitude test consisting of 1 Cycle in accordance with the following parameters as per MIL-STD-810F, Method 500.4 Altitude : 20Km Duration : 1hour soak Altitude change rate: 10 Meter/Sec (can be changed subject to chamber conditions) A4.5.1 OPERATIONAL TESTING Transfer Response test as per Para A4.1.1 to be carried out. Table-1 (PQT) <table><tr><th>SI</th><th>Test</th><th>Compliance (Y/N)</th><th>Signature (BEL Rep)</th><th>Signature (AQA)</th><th>Remarks</th></tr><tr><td>1</td><td>Random Vibration Test</td><td></td><td></td><td></td><td></td></tr><tr><td>2</td><td>Low temperature-Storage</td><td></td><td></td><td></td><td></td></tr><tr><td>3</td><td>Low temperature-Operational</td><td></td><td></td><td></td><td></td></tr><tr><td>4</td><td>High temperature-Storage</td><td></td><td></td><td></td><td></td></tr><tr><td>5</td><td>High temperature-Operational</td><td></td><td></td><td></td><td></td></tr><tr><td>6</td><td>Humidity (Non Operational)</td><td></td><td></td><td></td><td></td></tr><tr><td>7</td><td>Altitude (Operational)</td><td></td><td></td><td></td><td></td></tr></table>								SI	Test	Compliance (Y/N)	Signature (BEL Rep)	Signature (AQA)	Remarks	1	Random Vibration Test					2	Low temperature-Storage					3	Low temperature-Operational					4	High temperature-Storage					5	High temperature-Operational					6	Humidity (Non Operational)					7	Altitude (Operational)				
SI	Test	Compliance (Y/N)	Signature (BEL Rep)	Signature (AQA)	Remarks																																																		
1	Random Vibration Test																																																						
2	Low temperature-Storage																																																						
3	Low temperature-Operational																																																						
4	High temperature-Storage																																																						
5	High temperature-Operational																																																						
6	Humidity (Non Operational)																																																						
7	Altitude (Operational)																																																						

 भारत इलेक्ट्रॉनिक्स <i>BHARAT ELECTRONICS</i>		ACCEPTANCE TEST PROCEDURE			DOC CODE ATP	PART NUMBER 4556 109 201 15		
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ANNEXURE B

